

FROZEN SITE-PROBE ACCURACY IS NOT A STANDALONE HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level MRI harmonization is often judged with a site probe, a classifier that predicts the acquisition site: lower accuracy on harmonized images is read as better harmonization. We examine how reliable such verdicts are, starting from a frozen probe trained on unharmonized images, on raw ABIDE I T1-weighted MRI (1,112 subjects, 17 sites) with ten harmonized outputs as test cases. Verdicts depended on probe training: five probes trained with one recipe gave the same CycleGAN output balanced accuracies from 0.13 to 0.59, and a longer, stabilized recipe varied far less but found the site more decodable than the frozen probe in eight of ten outputs. The lowest accuracy, as low as probes trained on shuffled labels, came from an output that destroyed anatomy. Probes trained on harmonized images decoded the source site far better than raw-trained probes (+0.42 to +0.55 balanced accuracy, paired intervals above zero), also within the brain and from intensity histograms, so low frozen-probe accuracy does not establish removal of site-label information. The output with the smallest intensity-distribution change barely moved toward the target. We recommend several well-trained probes, including output-trained ones, against a shuffled-label baseline, with target alignment reported apart from image change.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Acquisition differences between sites confound pooled MRI analyses and degrade models on unseen scanners [1, 2]. Image-level harmonization translates scans toward a common appearance. Travelling subjects [3] or paired protocols [4] allow direct checks on few scanners, so large retrospective cohorts are judged indirectly [5]. A common proxy is a *site probe*: a classifier that predicts the acquisition site from an image, which is easy on raw MRI [2]. If its accuracy drops on harmonized images, site effects are taken to be removed [6, 7]; this is reported alongside similarity to the input, such as PSNR, and intensity-distribution distances. Some studies retrain the probe on harmonized images [6] or on image features [7, 8]. Others train it on the original images and apply it unchanged to the harmonized ones: the PRISM method [9], for example, reports this accuracy drop for two classifiers, with anatomy and segmentation checks, as evidence of harmonization. This frozen-probe protocol is our starting point.

Each proxy measures less than it appears to. A frozen probe reports what one classifier recovers; it can also fail because outputs

are unfamiliar, damaged or noisy. Removing an attribute until a fixed adversary fails does not stop a newly trained classifier from recovering it [10]. Site is also decodable at about 0.9 chance-corrected balanced accuracy from frozen, even randomly initialized, encoders, so the signal lies partly in low-level image statistics [11]. Similarity to the input rewards doing nothing, and small change says nothing about the target.

We examine how reliable site-probe verdicts are, starting from the frozen probe of an earlier benchmark, on outputs of seven harmonization families. Site classifiers retrained after ComBat [12] or histogram matching [13] still predict site; we quantify how probe training changes the verdict and whether low frozen-probe accuracy survives probes trained on harmonized outputs, separating geometry from intensity, measuring target alignment and including a destructive counterexample. We contribute a protocol with paired uncertainty and probe controls, these findings and a four-item reporting checklist (Sec. 5).¹ We neither propose nor rank harmonizers.

2. DATA AND AUDITED OUTPUTS

Cohort. Raw ABIDE I T1-weighted scans [14] of 1,112 subjects from 17 sites are split by subject, stratified by site, into 890 training, 113 validation and 109 test subjects. Volumes are reoriented, clipped to $[p_1, p_{99}]$ and scaled to $[0, 1]$; one axial slice per subject is chosen by foreground area, cropped to a dilated head mask and resized to 256×256 . NYU, the largest site, is the target; 19 test subjects are from NYU and 90 from 16 source sites. Exact ties made slice choice host-dependent, so slices are frozen and verified for every output. The label is coarser than the acquisition: ABIDE I releases UM, UCLA and Leuven as two sub-samples each, merged by our labels, and 11 of 17 sites hold more than one voxel size or matrix (UCLA includes 10 scans at $1.5 \times 1.5 \times 4$ mm); no per-subject scanner identifier is released.

Audited outputs. Earlier benchmark outputs were evaluated without retraining: NeuroCombat [1, 15] per pixel, fit on the test cohort itself (transductive); histogram matching of foreground quantiles to a pooled 17-site training reference [16]; CycleGAN [17]; StarGAN [18], aggressive and conservative; a DLEST-style disentangled baseline (content encoder and AdaIN site styles, without DLEST’s energy-based latent sampler [6]) after 1,000 and 1,500 steps; conditional diffusion image-to-image translation [19, 20] (strength 0.35, 50 DDIM steps), sampled twice from one checkpoint; and adapted volumetric HCLD [7]. This gives ten outputs; Table 1 reports the first diffusion draw. Learned methods were

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¹Code, the evaluation protocol with all amendments, and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

trained on the training split. Test subjects were inspected in the earlier benchmark, so this is a retrospective reanalysis; each audit experiment was written into a protocol before its outcome was seen.

3. AUDIT PROTOCOL

Change and target alignment. PSNR on a fixed $[0, 1]$ scale and normalized cross-correlation (XCorr) compare output with input on whole images; generators return NYU inputs almost unchanged, so we summarize the 90 source subjects. *Change* is the per-subject Wasserstein-1 distance W between input and output intensities. *Target alignment* is the Wasserstein-1 distance and KL divergence between foreground intensities (defined on the raw slice) and a reference from the 147 NYU training subjects; ΔW_{NYU} is the output's distance minus the input's.

Site probes. Image probes are single-channel ResNet-18s [21] predicting 17 sites, tested on the fixed slices; unless stated otherwise they use AdamW (learning rate 3×10^{-4} , batch 64, 500 steps) and random affine augmentation and are selected on validation BA. We train 44 in five families:

Frozen (1): the earlier benchmark's own checkpoint.

Benchmark recipe (5 seeds): that training repeated, on a random axial slice of a raw training volume at each step. Does the seed change the verdict?

Converged (5 seeds): four times the steps, cosine decay and reseeded data-loader workers (the benchmark recipe repeats random draws across workers), tested at the final epoch. Do they agree?

Shuffled labels (4): labels permuted between subjects. Is the probe reading images, not labels?

Slice probes (24, plus 6 restricted-input networks and 8 logistic fits): trained on one fixed slice per training subject, from raw images or one method's outputs, selected on validation and tested on test subjects. Is site still decodable? Restricted variants see only head silhouettes or brain masks (geometry), only foreground intensity histograms (intensity; logistic regression), or slices inside raw-volume HD-BET brain masks [22].

The three slice-probe methods were chosen in advance to span intensity, adversarial and diffusion families; the converged, silhouette, histogram and brain-only probes were added after earlier results but before their own.

Statistics. Source BA averages the 16 source sites (2–15 test subjects each). A uniform guess over the 17 classes scores $1/17 \approx 0.06$; shuffled-label probes, our empirical null, scored 0.03–0.08. Intervals are 95% percentiles of 2,000 within-site subject bootstrap replicates, paired across methods and probes, and conditional on these sites and checkpoints.

4. RESULTS

Probe validity. Raw source BA was 0.95 for the frozen probe, 0.81–0.98 for benchmark-recipe and 1.00 for converged probes. Probes learn site-related structure, not only scanner appearance: head silhouettes alone gave 0.73–0.86.

Verdicts depend on probe training. Benchmark-recipe probes assigned very different site BA to the same outputs (Table 1): CycleGAN 0.13–0.59 and aggressive StarGAN 0.18–0.54, with Kendall τ 0.51–0.87 between method orderings. Their validation BA fluctuated between epochs (0.88, 0.50, 0.87), and raw accuracy did not reveal it: the probe with raw BA 0.98 gave CycleGAN 0.20. Much of this spread was a training-stability effect: converged probes (validation BA within 0.035 over the last five epochs) varied far less

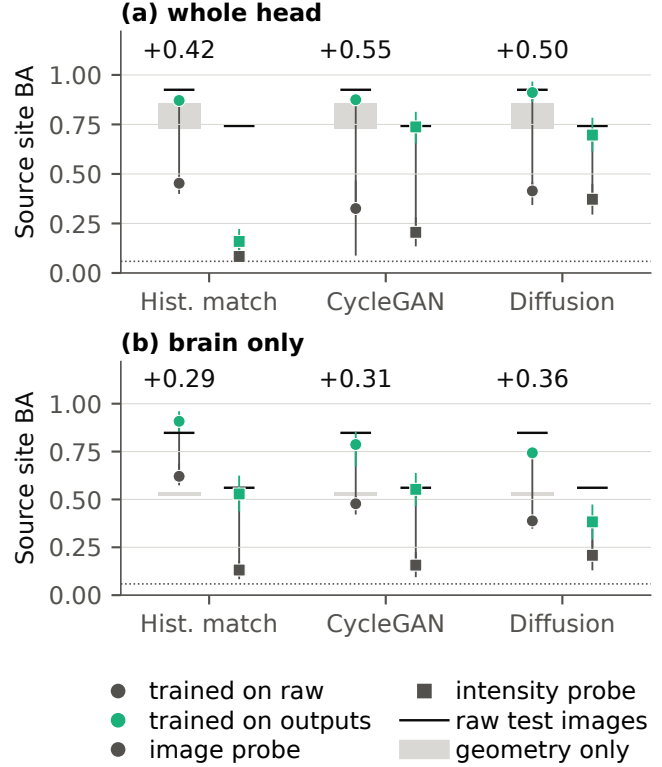


Fig. 1. Probes trained on raw slices (grey) or on a method's outputs (aqua), tested on that method's outputs (diffusion: second draw), on (a) whole heads and (b) brain-only slices: image probes (circles, mean and range over 3 seeds) and intensity-histogram probes (squares, 95% interval). Numbers: paired difference in three-seed mean BA between output- and raw-trained image probes. Black bar: raw-trained probe on raw test slices. Grey: image probes given only head silhouettes or brain masks. Dotted line: uniform chance.

across seeds (τ 0.82–0.96; per-output ranges at most 0.16), but every converged probe exceeded the frozen probe's interval on all outputs except adapted HCLD and NeuroCombat, e.g. aggressive StarGAN 0.65–0.81 against 0.21. Method orderings changed (τ 0.64–0.73 with the frozen probe).

The lowest site accuracy came from destruction. The adapted HCLD output reached BA 0.07 [0.06, 0.08] under the frozen probe and 0.06–0.11 under every benchmark-recipe and converged probe, near the shuffled-label baseline, with the lowest PSNR (13.9 dB) and XCorr (0.71), loss of anatomy (Fig. 2), and intensities moved away from NYU ($\Delta W_{\text{NYU}} = +0.108$). On this output the frozen probe labelled 83 of its 84 errors as Yale, not NYU, whereas 65–80% of errors on CycleGAN, aggressive StarGAN and diffusion outputs named NYU. It is a counterexample for the metric, not a verdict on published HCLD.

Small change is not target alignment. Diffusion changed the intensity distribution least ($W = 0.008$) yet barely moved it toward NYU ($\Delta W_{\text{NYU}} = -0.0013$ [−0.0023, −0.0004], 1% of the raw distance) and raised KL divergence to NYU from 0.30 to 0.83. Outputs are also a random draw: resampling the checkpoint changed images more than harmonization did (per-subject mean absolute foreground difference, median 0.085 against 0.075), at un-

Table 1. Source subjects ($n = 90$). PSNR and XCorr compare output with input; ΔW_{NYU} : change in Wasserstein-1 distance to NYU (negative is closer). Site BA: frozen probe with 95% interval; five benchmark-recipe and five converged probes as mean [min, max]. Uniform chance 0.06; shuffled-label probes 0.03–0.08. [†]Fit on the test cohort. [‡]Pooled 17-site reference.

Method	Change and target alignment			Source-site balanced accuracy of probes trained on raw images		
	PSNR	XCorr	ΔW_{NYU}	Frozen probe	Benchmark recipe	Converged recipe
Raw input	–	–	0	0.95	0.90 [0.81, 0.98]	1.00 [1.00, 1.00]
NeuroCombat [†]	20.0	0.940	+0.003	0.27 [0.21, 0.32]	0.33 [0.26, 0.37]	0.34 [0.28, 0.42]
Histogram matching [‡]	23.0	0.982	-0.053	0.44 [0.37, 0.52]	0.44 [0.27, 0.59]	0.61 [0.55, 0.68]
CycleGAN	21.3	0.977	-0.036	0.31 [0.25, 0.37]	0.28 [0.13, 0.59]	0.49 [0.44, 0.54]
StarGAN (aggr.)	16.0	0.896	-0.019	0.21 [0.15, 0.27]	0.35 [0.18, 0.54]	0.70 [0.65, 0.81]
StarGAN (cons.)	26.9	0.991	-0.021	0.83 [0.79, 0.86]	0.78 [0.66, 0.89]	0.98 [0.94, 0.99]
DLEST-style 1500	26.6	0.993	-0.020	0.67 [0.61, 0.72]	0.73 [0.65, 0.82]	0.96 [0.95, 0.97]
DLEST-style 1000	28.4	0.993	-0.009	0.71 [0.65, 0.77]	0.84 [0.76, 0.89]	0.99 [0.96, 1.00]
Diffusion img2img	21.7	0.948	-0.001	0.37 [0.32, 0.42]	0.42 [0.24, 0.63]	0.57 [0.52, 0.59]
Adapted HCLD	13.9	0.706	+0.108	0.07 [0.06, 0.08]	0.08 [0.06, 0.11]	0.08 [0.06, 0.10]

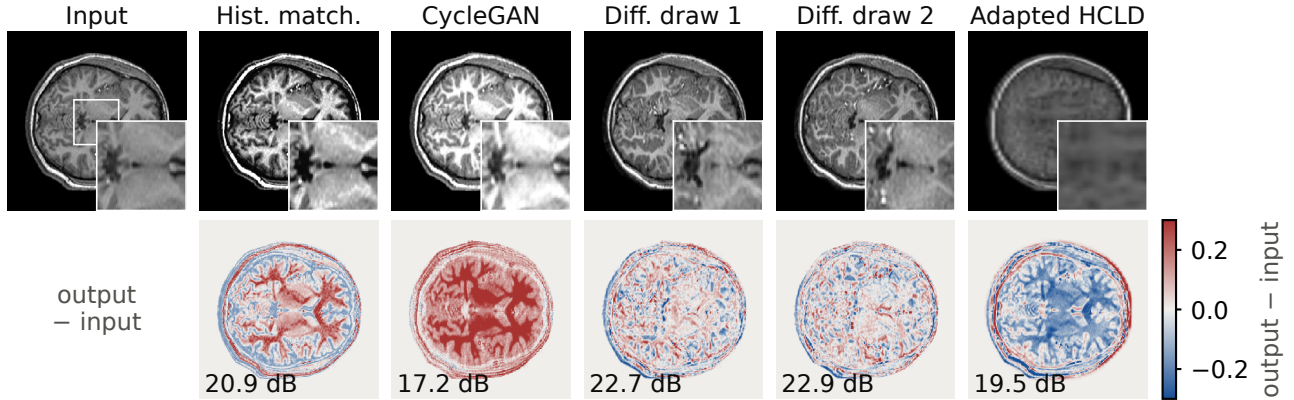


Fig. 2. Source test subject with median PSNR over nine outputs (all but DLEST-style 1500); insets are a $2\times$ zoom of the boxed region. Bottom: output minus input, with this subject’s PSNR. The adapted HCLD output loses anatomy; diffusion adds texture that differs between draws.

changed PSNR.

Is site still decodable? Slice probes trained on raw slices reached BA 0.92 on raw test slices but 0.45, 0.33 and 0.41 on histogram-matched, CycleGAN and diffusion outputs, a drop interpreted as evidence of harmonization [9]. Probes trained on each method’s outputs recovered the source site from its test outputs with BA 0.87, 0.87 and 0.91 (Fig. 1a), exceeding raw-trained probes by +0.42 [0.36, 0.48], +0.55 [0.52, 0.58] and +0.50 [0.45, 0.54] (paired differences of three-seed mean BA). These probes use the shorter benchmark recipe (Table 1), so they may understate decodability. Because head geometry identifies sites, we repeated this on brain-only slices: raw-trained probes reached 0.85 on raw brains but 0.62, 0.48 and 0.39 on the outputs, and probes trained on outputs 0.91, 0.79 and 0.74 (Fig. 1b; differences +0.29 to +0.36, all intervals above zero, also with final-epoch checkpoints); brain masks alone gave 0.52–0.54. Intensity histograms discard spatial arrangement. On whole heads, an output-trained histogram probe recovered little from histogram matching (0.16 [0.11, 0.22]), whose head histograms are matched by construction, but recovered CycleGAN (0.74) and diffusion (0.70) as well as raw ones (0.74). Within the brain it recovered all three: 0.53 [0.44, 0.62], 0.55 [0.46, 0.64] and 0.38 [0.29, 0.47], against 0.56 on raw brains and 0.13–0.21 for the raw-trained

probe. Low frozen-probe accuracy does not establish removal of site-label information, much of which output-trained probes decode.

5. DISCUSSION AND RECOMMENDATIONS

One site number is not a verdict: it moved by up to 0.46 with the probe seed and was lowest for the most destructive output. We recommend reporting (1) several converged probes, their recipe, raw accuracy, a shuffled-label null and a fidelity measure; (2) probes trained on the outputs, with channel-restricted and geometry controls, whose high accuracy shows retained site-label information but not, by itself, failed harmonization, as head and brain shape also predict site; (3) intensity alignment to the target apart from change, on source subjects only: the output with the smallest change closed 1% of its distance to NYU; and (4) paired uncertainty over subjects and generator draws.

Limitations. We use one cohort, one target site and one axial slice per subject, so volumetric effects are not assessed. Without anatomical ground truth, PSNR and XCorr measure agreement with the input, not correctness. Brain masks leave brain shape, slice level and orientation, and site is confounded with demographics and with acquisition geometry that varies within a site; our controls bound

these channels but cannot separate scanner effects from biological differences. Generators were not retuned, and one baseline is transductive.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by ABIDE I [14], whose sites shared the fully anonymized data with local ethics approval or an explicit waiver. Ethical approval was not required as confirmed by the license attached with the open access data.

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